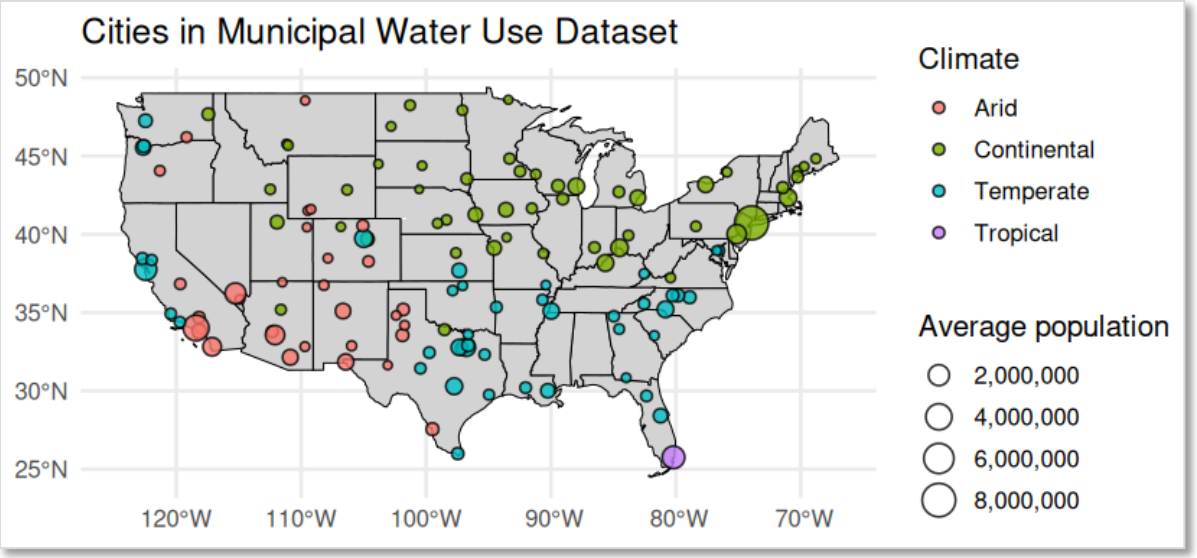
Relationship between population growth and residential per capita water use in 126 U.S. cities

Viviane Baji
EDS Capstone Slide
2026.03.30
Mentor: Eliana Stone



Datasets and variables

1) Municipal water use (2005-2017)*
- City/Census ID
- Monthly residential billed gallons
- Population (yearly)

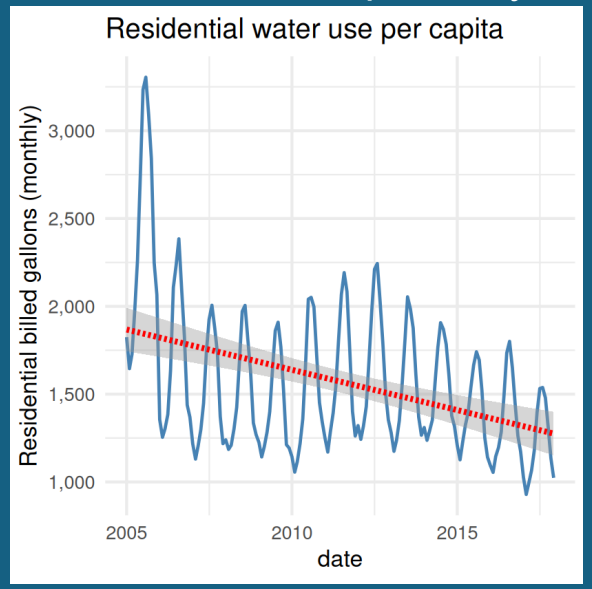
2) City characteristics*
- City/Census ID
- Climate region
- Population
- Other (not used)

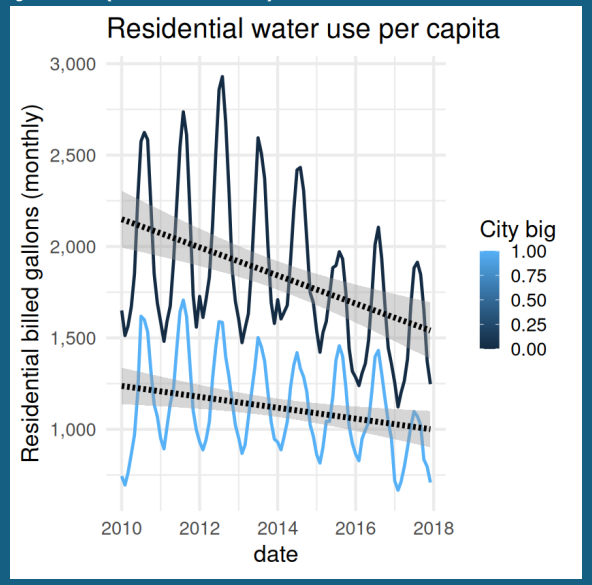
3) City location shapefile*
- Point data
- City/Census ID

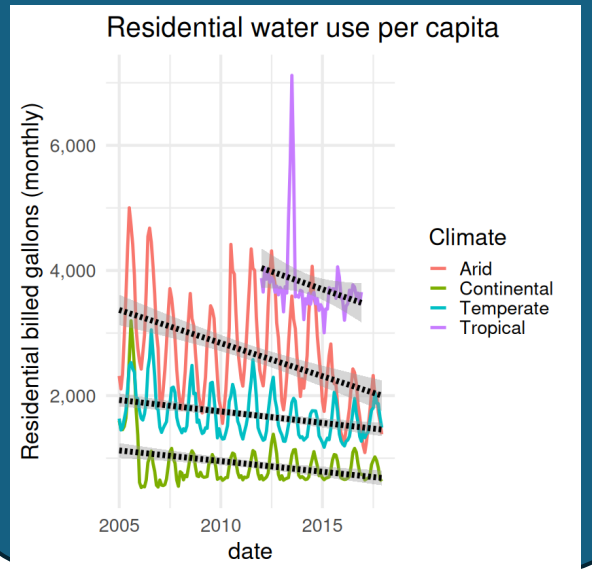
Data Cleaning
- Missingness
- New columns
- Joining

**assembled by Chinnasamy et al., 2021*

Exploratory Analysis (all cities)







Explanatory Analysis (Regression)

```
OLS estimation, Dep. Var.: log(res_billed_gallons)
Observations: 13,258
Standard-errors: IID
                 Estimate Std. Error   t value  Pr(>|t|)
(Intercept)      19.15374   0.011896 1610.1086 < 2.2e-16 ***
city_big_growth   2.17669   0.035946   60.5539 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.29246   Adj. R2: 0.216618
```